Flows and RealNVP

Recall the following result about change of variable: Suppose $z \sim \pi(z)$ and $f(x) = z$ for some invertible f. The distribution of $x \sim p(x)$ can be determined by

$$\begin{aligned} p(x) &= \pi(z) \left| \frac{dz}{dx} \right| \\ &= \pi(f(x)) |f^{'}(x)|. \end{aligned}$$

Its multivariate version is

$$p(x) = \pi(f(x)) \left| \det \left(\frac{\partial f}{\partial x} \right) \right|,$$

where $\frac{\partial f}{\partial x}$ is the Jocabian matrix.
Suppose data $x \sim p(x)$ and latent variable $z \sim \pi(z)$. **Normalizing Flow Model** uses a series of invertible maps, $f_1, ..., f_K$, with

$$x = z_0 \xrightarrow{f_1} z_1 \xrightarrow{f_2} z_2 \xrightarrow{f_3} ... \xrightarrow{f_K} z_K = z.$$

Suppose z_i has distribution p_i. We have

$$p_i(z_i) = p_{i+1}(f_{i+1}(z_i)) \left| \det \left(\frac{\partial f_{i+1}}{\partial z_i} \right) \right|.$$

Hence, if we put $F = f_K \circ ... \circ f_1$ mapping x to z,

$$\begin{aligned} p(x) = p_0(z_0) = p_K(F(z_0)) &\prod_{i=0}^{K-1} \left| \det \left(\frac{\partial f_{i+1}}{\partial z_i} \right) \right| \\ = \pi(F(x)) &\prod_{i=0}^{K-1} \left| \det \left(\frac{\partial f_{i+1}}{\partial z_i} \right) \right|. \end{aligned}$$

The log-likelihood therefore is

$$\log p(x) = \log \pi(F(x)) + \sum_{i=0}^{K-1} \log \left| \det \left(\frac{\partial f_{i+1}}{\partial z_i} \right) \right|.$$

Necessary conditions of the design of the invertible maps $f_1, ..., f_K$ include the following:
(1) they are easy to compute;
(2) the determinants of the Jocobians are easy to compute;
(3) their inverses are easy to compute.

Real NVP (Real-valued Non-Volume Preserving) uses invertible maps satisfying the requirements above. Suppose data x and latent variable z are in $\mathbb{R}^D$. Let $1 \leq d < D$ and $s, t : \mathbb{R}^d \to \mathbb{R}^{D-d}$ be two neural networks. Define the map f from x to z by

$$\begin{aligned} f &: x \to z \\ z_{1:d} &= x_{1:d} \\ z_{d+1:D} &= x_{d+1:D} \odot \exp(s(x_{1:d})) + t(x_{1:d}), \end{aligned}$$

where $\odot$ is element-wise product.
The inverse is easy to compute:

$$\begin{aligned} x_{1:d} &= z_{1:d} \\ x_{d+1:D} &= (z_{d+1:D} - t(z_{1:d})) \odot \exp(-s(z_{1:d})). \end{aligned}$$

The Jocobian matrix is

$$\begin{bmatrix} I_d & O_{d\times(D-d)} \\ * & \mathrm{diag}(\exp(s(x_{1:d}))) \end{bmatrix},$$

and its determinant is

$$\exp\left(\sum_{j=1}^{D-d} s(x_{1:d})_j\right).$$

We also note that the partitioning can be programmed using a binary mask $b \in \{0,1\}^D$:

$$z = b \odot x + (1-b) \odot (x \odot \exp(s(b \odot x)) + t(b \odot x)).$$

The inverse is

$$x = b \odot z + (1-b) \odot (z - t(b \odot z)) \odot \exp(-s(b \odot z)).$$